Major Advanced Topics

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Strong-Field Ionization Phenomena Revealed by Quantum Trajectories

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Laser-atom interaction in Strong field

- The Keldysh parameter

$$\gamma_{\mathrm{K}} \equiv \omega\sqrt{2I_p}/E_0$$

$\gamma_{\mathrm{K}} < 1 \longrightarrow$ Tunnel ionization (horizontal channel)
$\gamma_{\mathrm{K}} > 1 \longrightarrow$ Multiphoton ionization (vertical channel)
$\gamma_{\mathrm{K}} \sim 1 \longrightarrow$ Intermediate regime

  - Provides a quantitative understanding of the dominant photoionization mechanism

  - Disregards the nature of the binding potential and excited states
  - Reduces the intricacies of the field to the field strength only
  - Wrongly assumes that the tunneling wavepacket always start in the classically allowed region
  - Over the barrier ionization (OBI) is poorly rendered

- Since the dynamics of the electron becomes classical, trajectories based approaches are appropriate

- This paper attempts at

  1. proposing a more suitable parameter than γ_{K} for the description of the interaction
  2. using the above, quantify the efficiency of OBI

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The de Broglie-Bohm quantum theory

Trajectories are researched by solving the TDSE, first

$$i\frac{d}{dt}\Psi(t) = H(t)\Psi(t), \qquad H(t) = H_0 - \boldsymbol{p}\boldsymbol{A}(t), \qquad \boldsymbol{A}(t) = -A_0\cos^4\left(\frac{\pi t}{NT}\right)\sin(\omega t)\hat{\boldsymbol{x}},$$

$$\boldsymbol{E}(t) = -\frac{d}{dt}\boldsymbol{A}(t), \qquad t \in \left[-NT/2, NT/2\right], \qquad T = 2\pi/\omega,$$

- Bohmian formalism and quantum potential

$$i\frac{\partial\psi}{\partial t} = \left(-\frac{1}{2}\nabla^2 + V\right)\psi, \quad \text{with} \quad \psi = R\exp(iS), \qquad \rho(\boldsymbol{r},t) = |\psi(\boldsymbol{r},t)|^2$$

Using the above expression of the wavefunction ψ in the Schrödinger equation, we get leaves us with

$$\frac{\partial R}{\partial t} = -\frac{1}{2}\left\{R\nabla^2 S + 2\nabla R\nabla S\right\},$$

$$\frac{\partial S}{\partial t} = -\frac{1}{2}\left\{(\nabla S)^2 + \left(\frac{\nabla^2 R}{R}\right)\right\} - V \qquad Q = -\frac{1}{2}\rho^{-1/2}\nabla^2\rho^{1/2}$$

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Bohmian theory - Velocity field and trajectories

- In the Bohmian formalism, velocity is given by

$$\boldsymbol{v}(\boldsymbol{r},t) = \boldsymbol{j}(\boldsymbol{r},t)/\rho(\boldsymbol{r},t), \qquad \boldsymbol{j}(\boldsymbol{r},t) = Re\left(\Psi^*(\boldsymbol{r},t)\left(-i\Delta + \boldsymbol{A}(t)\right)\Psi(\boldsymbol{r},t)\right)$$

  - Retrieving ψ from the TDSE
  - Propagation of the trajectories from different positions $\boldsymbol{r}_0$
  - Reconstruction of the wavefunction from the trajectories at any given time

- Computation of time and radii exit t_{ex} and r_{ex} from the condition

$$\Delta\mathcal{E}(t_{\text{ex}}) = 0, \qquad \Delta\mathcal{E}(t) = v^2(t)/2 + Q(t),$$

  where $\Delta\mathcal{E}(t)$ is the difference between the energy of the trajectory $\mathcal{E}(t)$ and I_p

- The quantum potential $Q(t)$ is responsible for all quantum effects and the nonzero velocity at the exit

- In the classical limit

$$Q(t) \longrightarrow 0$$

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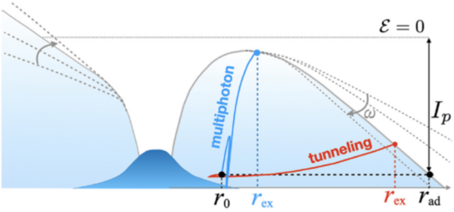
The tunneling parameter

- For ionizing trajectories, asymptotic behavior is such as $\mathcal{E}(\infty) \geqslant 0$
- The adiabatic radius r_{ad} corresponds to a trajectory such as $\mathcal{E} = -I_p$

$$\beta\left(\boldsymbol{r}_0\right) \equiv \frac{r_{\text{ex}}\left(\boldsymbol{r}_0\right) - r_0}{r_{\text{ad}}\left(\boldsymbol{r}_0\right) - r_0}$$



- $0 \leqslant \beta \leqslant 1$ quantifies the degree of tunneling for a trajectory starting at r_0
- For vertical ionization, $r_{\text{ex}} \longrightarrow r_0 \Longrightarrow \beta \longrightarrow 0$
- For horizontal ionization, $r_{\text{ex}} \longrightarrow r_{\text{ad}} \Longrightarrow \beta \longrightarrow 1$
- β aslo exists in the eventuality that $r_{\text{ex}} \geqslant r_{\text{ad}}$ (OBI...)

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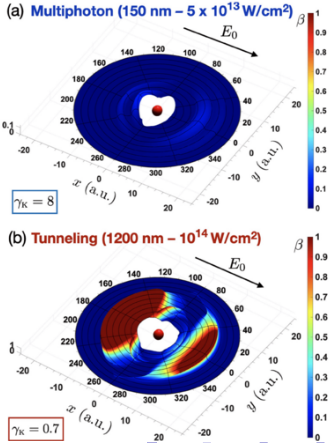
Photoionization of atomic hydrogen as a function of r_0

- 2-dimensional plot of $\beta(\boldsymbol{r}_0)$ as a function of r_0 in the $xy-$plane
- 2 different pulses describing MPI and TI are used
- Fig. 2(b) provides additional details on the dynamics of the ionization
  - Tunnel ionization dominates in the direction longitudinal to the field past a certain r_0
  - Multiphoton mechanism occurs in the direction transversal to the field ionization
  - At large distances, the mechanism naturally exhibits a multiphoton character



$\beta \longrightarrow 0$	vertical channel
$\beta \longrightarrow 1$	horizontal channel

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Quantum adiabatic parameter

- Statistically averaged tunneling level of the ionization process

$$\bar{\beta} = W_0^{-1} \int_{\mathcal{D}} \beta\left(\boldsymbol{r}_0\right) \rho\left(\boldsymbol{r}_0, 0\right) d^3 \boldsymbol{r}_0, \qquad \rho(\boldsymbol{r}_0, 0) = |\psi(\boldsymbol{r}, 0)|^2 .$$

  - $\mathcal{D}$ denotes the set of ionizing trajectories
  - $W_0 = \int_{\mathcal{D}} \rho\left(\boldsymbol{r}_0, 0\right) d^3 \boldsymbol{r}_0$ is the total ionization probability

- To compare with the Keldysh parameter γ_{K}, they define the quantum adiabatic parameter γ_q

$$\gamma_q \equiv \frac{1 - \bar{\beta}}{\bar{\beta}}$$

  - γ_q has the same limits a nd meaning as γ_{K}
  - $\gamma_q \longrightarrow 0$ for TI and $\gamma_q \longrightarrow \infty$ for MPI
  - $\gamma_q \sim 1$ for equal TI and MPI efficiency, we would then have $\bar{\beta} = 1/2$

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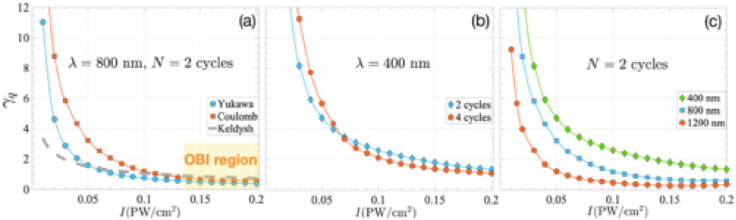
Quantum adiabatic parameter as a function of the intensity



- Fig. 3(a) Comparison of γ_q and γ_K for different intensities and different potentials
  - The difference between the potentials is sharp at low intensities. γ_q is more sensitive than γ_K
  - Accross different potentials, while γ_K remains unaffected, γ_q reveals the peculiarities of the structure
- Fig. 3(b) Comparison of γ_q for different pulse durations
  - 4 cycles has more enhanced MPI and TI mechanisms because more laser peaks
- Fig. 3(c) Comparison of γ_q for different wavelengths
  - The results agree with general prediction

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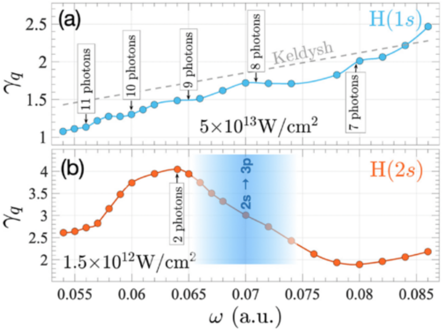
Quantum adiabatic parameter as a function of the frequency

- Fig. 4(a) While γ_K shows linearity, γ_q shows nonuniformity
  - Correspond to channel closings known to suppress ionization
  - m-photon channel closure is realized when

$$m\omega \leqslant I_p + U_p, \qquad U_p = E_0^2/(4\omega^2).$$

- Fig. 4(b) Results for 2 photon ionization in the $2s$ state
  - Sharp drop near resonnance indicating a preference for TI
  - Population transfer to $3p$ instead of Rydberg states favors TI
  - to $3p$ maintain the e^- within reach of a manageable horizontal ionization.



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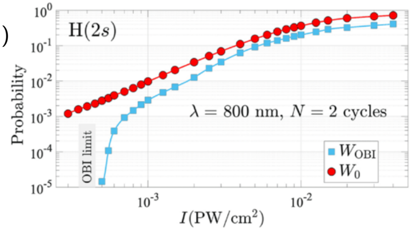
Over the barrier ionization via quantum trajectories

- OBI happens if

$$\mathcal{E}(t) > I_p \qquad \text{or} \qquad \mathcal{E}(t_{\mathrm{ex}}) \leqslant I_p \tag{2}$$

- W_{OBI} is the OBI ionization probability W_0 is the ionization probability
- Going by this definition, Fig. 5 compares W_0 and W_{OBI} for different intensities
  - W_{OBI} experiences a sharp increase when the top of the barrier becomes lower than I_p
  - Subsequentially, the increase rate is similar to W_0
- At intensities about 10 times higher than the OBI limit, a steady state is reached where $W_{\mathrm{OBI}}/W_0 \sim 0.6$



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Closure

- Quantum trajectories represent a powerful alternative tool to investigate electron dynamics beneath and above the potential barrier
- The tunneling parameter $\bar{\beta}$ allows a mapping of the angular-resolved tunneling dynamics
- The quantum adiabatic parameterγ_q is superior to the Keldysh parameter γ_{K} in the sense that
  - Rare quantitative information can be probed about various high intensity effects
  - Importance of long range electronic potential and nonadiabatic effects are better rendered
- This might pave the way to a better understanding of other routinely studied dynamical effects (HHG, etc..)